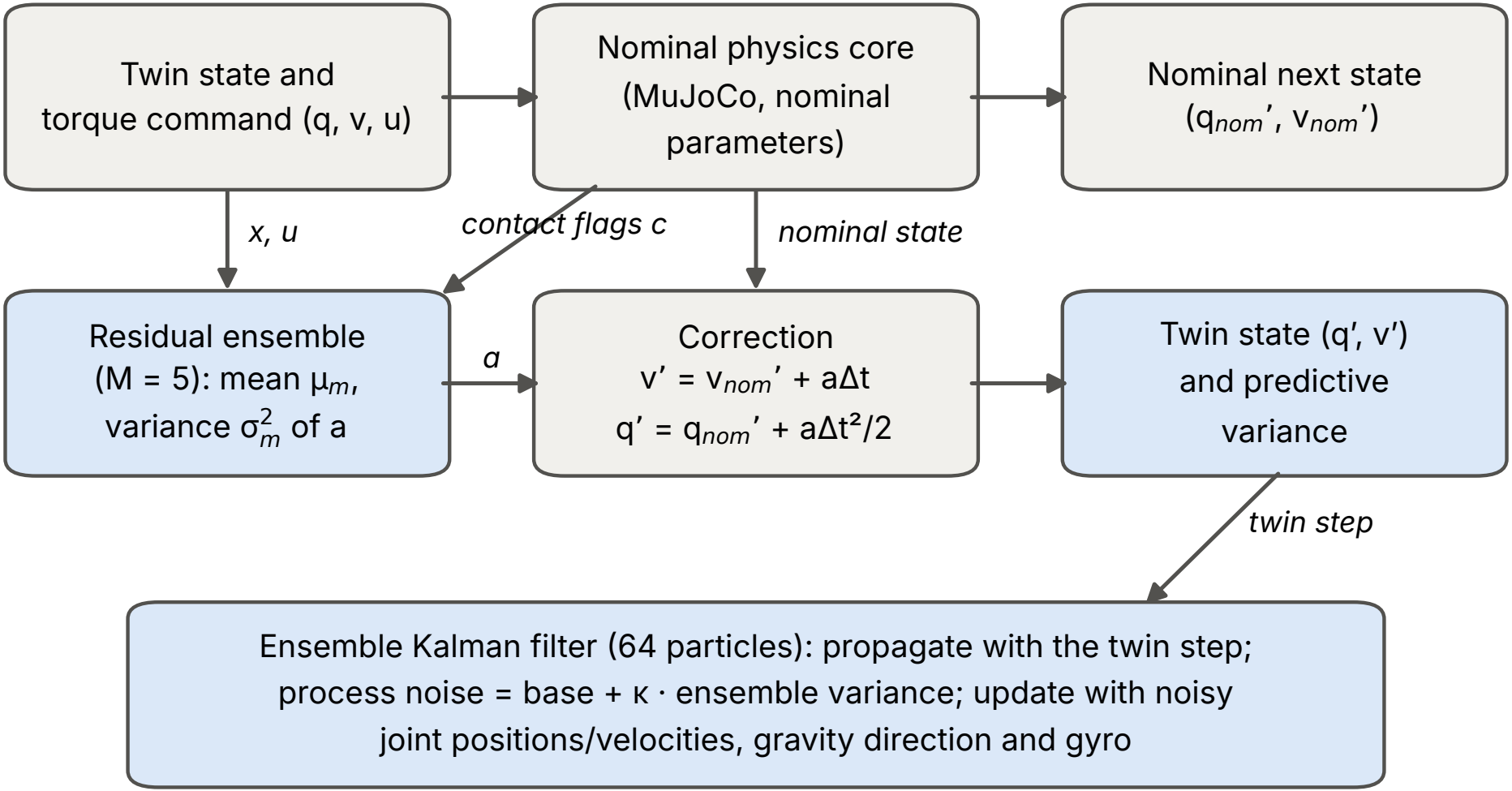
Twin state and torque command (q, v, u)
Nominal physics core (MuJoCo, nominal parameters)
Nominal next state (q_{nom}', v_{nom}')
x, u
contact flags c
nominal state
Residual ensemble (M = 5): mean μ_m, variance σ_m^2 of a
a
Correction
$v' = v_{nom}' + a\Delta t$
$q' = q_{nom}' + a\Delta t^2/2$
Twin state (q′, v′) and predictive variance
twin step
Ensemble Kalman filter (64 particles): propagate with the twin step; process noise = base + κ · ensemble variance; update with noisy joint positions/velocities, gravity direction and gyro